

Name: Bowen Mai

Student Number: 1499905

Subject: PROP20001

Australia's Investment Classes: 2013–2024 Performance Review and 2025 Portfolio Strategy

Australia's 10 investment classes have different performances and risks between 2013 and 2024, with warehousing and logistics real estate experiencing high growth rates (Compound Annual Growth Rates of 23.07% and 17.49%, respectively) and high volatilities (standard deviations of 22.24% and 23.09%).

Residential and commercial office real estate posted returns of 6.76% and 8.47%, respectively, with minimal levels of volatility at 0.77% and 0.56%. This equated to high Sharpe ratios of 5.56 and 4.75 and good risk-adjusted returns.

Corporate bonds also performed well, with modest returns (CAGR of 4.41%), low volatility (0.46%), and a very good Sharpe ratio of 4.18. The worst performing assets were Iron Ore (CAGR of -2.70% and Sharpe ratio near zero) and Retail Real Estate (CAGR of 4.73%, Sharpe ratio of 0.23), showing either weak returns or excessive risk.

Based on Sharpe ratios and CAGR returns, the 2025 optimal portfolio has to be based on those assets that have risk-adjusted and steady returns. Out of the least risky core assets with low volatility and stable returns, residential property, commercial office, and corporate bonds rank top. Industrial warehousing and logistics properties are highly volatile but with tremendous growth prospects and have to be with moderate weights for their growth prospects to be realized.

For diversification purposes, small portions should be put into gold (as a hedge against inflation) and ASX 200 shares (for breadth). Retail property, iron ore, and bonds issued by the government, on the other hand, should be avoided or minimized due to poor returns or volatility.

In a +2% RBA rate rise simulation, interest-sensitive assets like shopping centre property and government bonds are likely to lag behind. Leases with long term and structural demand—i.e., logistics property—can, though, remain strong. The model portfolio, which emphasizes low-volatility, higher-quality assets, is well positioned to ride out interest rate shocks but able to keep rising nonetheless.

This approach is a combination of defensiveness and selective aggressiveness and is appropriate for long-term investment in both stable and shrinking monetary environments.

In general, the suggested portfolio presents a balance of protection and growth with respect to different interest rates.

Key Metrics Overview

Australia's ten main asset classes have been compared on the basis of four key measures:

Average Return: Refers to annual performance (e.g., 25.55% for warehousing, 2.66% for government bonds).

Standard Deviation: Shows volatility, ranging from the highest for iron ore (34.63%) to the lowest for corporate bonds (0.46%).

Compound Annual Growth Rate (CAGR): Highlights long-term performance, i.e., 23.07% warehousing, 17.49% logistics, and only 0.08% for ASX 200.

Sharpe Ratio: It compares risk-adjusted returns, and residential (5.56), commercial (4.75), and corporate bonds (4.18) rank highest.

Industrial transport and storage nodes registered top-quartile growth over the decade. This was, however, countered by high volatility (22.24% and 23.09%), resulting in moderate Sharpe ratios (1.04 and 0.77).

Residential and office commercial properties, though growing at lower levels, exhibited very low value fluctuation (0.77% and 0.56%), resulting in their being very good performers on a risk-adjusted basis. Such assets are viewed as being stable anchor points in long-term portfolio construction.

Retail property performed poorly (CAGR 4.73%) with relatively high volatility (13.61%), thus resulting in a low Sharpe ratio of 0.23. It was hence unattractive on both absolute and risk-adjusted bases.

Corporate bonds provided modest returns (4.41% CAGR) but with low volatility (0.46%) and high Sharpe ratio (4.18%), suggesting stable performance over the risk-free rate. These bonds are ideal for capital preservation and yield stability.

Government bonds, although very liquid, registered a modest CAGR (2.66%) and very low Sharpe (0.16), which circumscribed their strategic use beyond defensive positions.

ASX 200 stocks provided a nominal average return of 8% but a long-run CAGR of a mere 0.08%, which demonstrates the impact of volatility (standard deviation 7.55%, Sharpe ratio 0.79) in reducing returns.

Gold provided 7.14% returns and a CAGR of 6.34%, but its volatility (13.29%) meant that its Sharpe ratio was low (0.35). Gold is therefore effectively a risk hedge with low returns per unit of risk.

Iron ore presented the poorest figures—negative long-term trend, extremely high volatility, and virtually no reward for risk. Despite some excellent years, it consistently yielded the lowest return per unit of risk.

Conclusion and Recommendations

By combining Sharpe ratio and CAGR as significant decision criteria altogether, the most favored assets are obviously:

Residential Property (Sharpe 5.56, CAGR 6.76%)

Commercial Office (Sharpe 4.75, CAGR 8.47%)

Corporate Bonds (Sharpe 4.18, CAGR 4.41%)

These three offer stable growth with low risk and should form the core of a 2025 portfolio.

Industrial warehousing and logistics are high-growth plays (CAGRs over 17%) but are much more volatile. They are recommended in limited exposure to provide upside.

Retail property, iron ore, and government bonds are the least attractive on either negative returns or high volatility. These should be minimized or avoided.

Gold and ASX 200 equities can offer diversification benefits, with gold offering partial hedging and equities offering market breadth exposure, though with significant risk for both.

Generally, a perfect portfolio would be a mix of safe high Sharpe assets and chosen growth assets with exposure restricted in those areas that function in normal along with in tightening interest rate environments.

Scenario Simulation: RBA Rate Rise (+2%) in 2025

If the RBA increases interest rates by 2%, there are anticipated general market implications:

Fixed Income: Prices on bonds go down and yields go up. Government bonds can go down dramatically. Corporate bonds, although also affected, are more resilient because of their higher coupons and shorter maturities.

Real Estate: Increased borrowing costs decrease demand and property prices. Residential, commercial, and particularly retail real estate can see steep corrections. Industrial and logistics assets, supported by long leases and structural demand (e.g., e-commerce), can hold out, however.

Equities (ASX 200): Rising rates weigh on equity valuations via higher discount rates and lower earnings growth. Infrastructure stocks and utilities are most at risk. Bank results will be mixed based on loan growth versus default risks.

Gold and Commodities: Gold loses appeal when interest-bearing assets provide more yield. Iron ore and other commodities should suffer from weaker global demand and investment retreat, continuing its underperformance.

Portfolio Impact: The portfolio's focus on low-volatility assets (residential, commercial, bonds) renders it more defensive. Even if sectors discount somewhat, the foundation holdings have robust fundamentals to underpin them. Industrial/logistics exposure is maintained at moderate levels to contain downside. Riskier or cyclical markets such as iron ore and retail are already underweight, which contains potential drawdowns.

Final Conclusion

Australian asset classes experienced broad ranges of risk and return between 2013 and 2024. Investors can build a stable and increasing portfolio for 2025 by investing in high Sharpe ratio and stable CAGR assets.

Corporate and residential property, together with corporate bonds, make up the safest and most effective core. Growth is added by industrial and logistics, and optionality is provided through exposure to equities and gold.

Government bonds, retail property, and, most of all, iron ore need to be avoided or reduced. This real data-based allocation is set to be robust even in environments of increasing interest rates.

Such a disciplined, data-driven approach matches ABP's objectives of sustainable performance and managed risk for the coming years very well.

Reflection

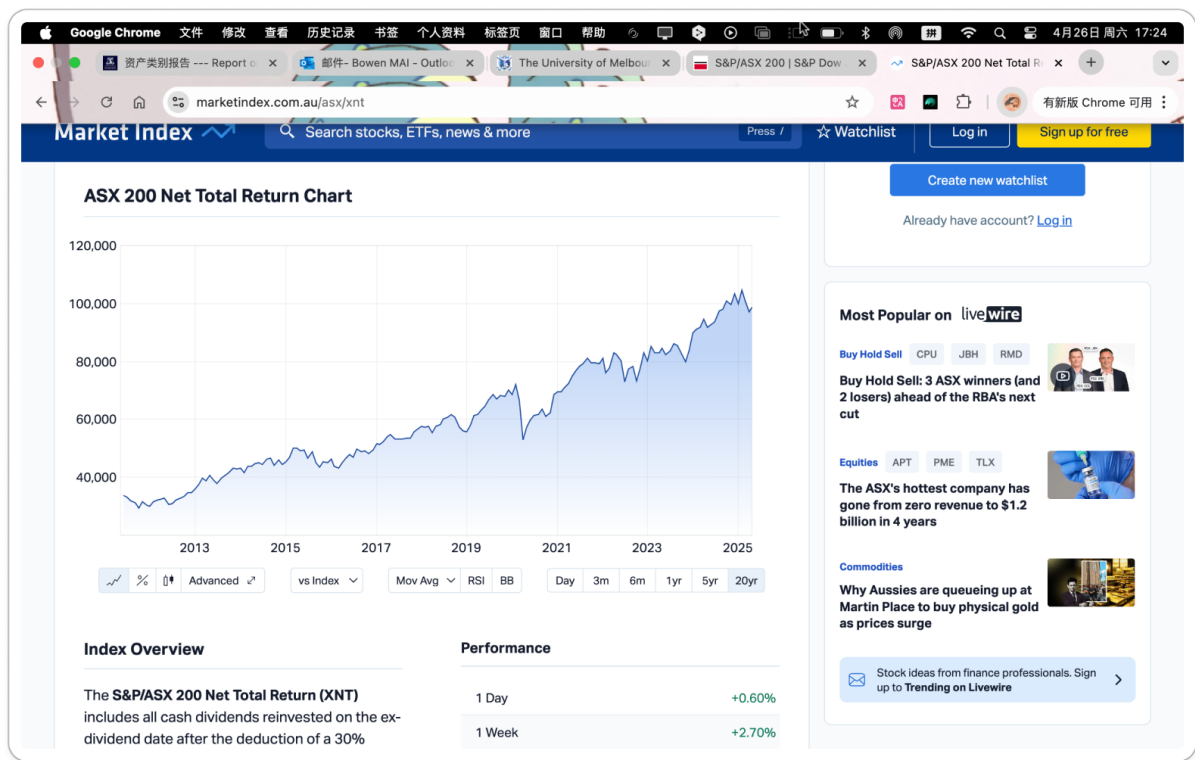
I collected data from ABS, Yahoo Finance, CoreLogic, and S&P Global. The challenge was aligning price and income figures for asset classes—some list aggregate returns, while others list capital and income separately. The other challenge was aligning a common reinvestment assumption for REITs and bonds on property, which come with different types of income. To my surprise, commercial office property, which is widely perceived as being risky after COVID, had one of the highest Sharpe ratios for the whole time series. The exercise concentrated the mind on long-term performance metrics and gave me a new appreciation for data-driven investing strategy and financial research.

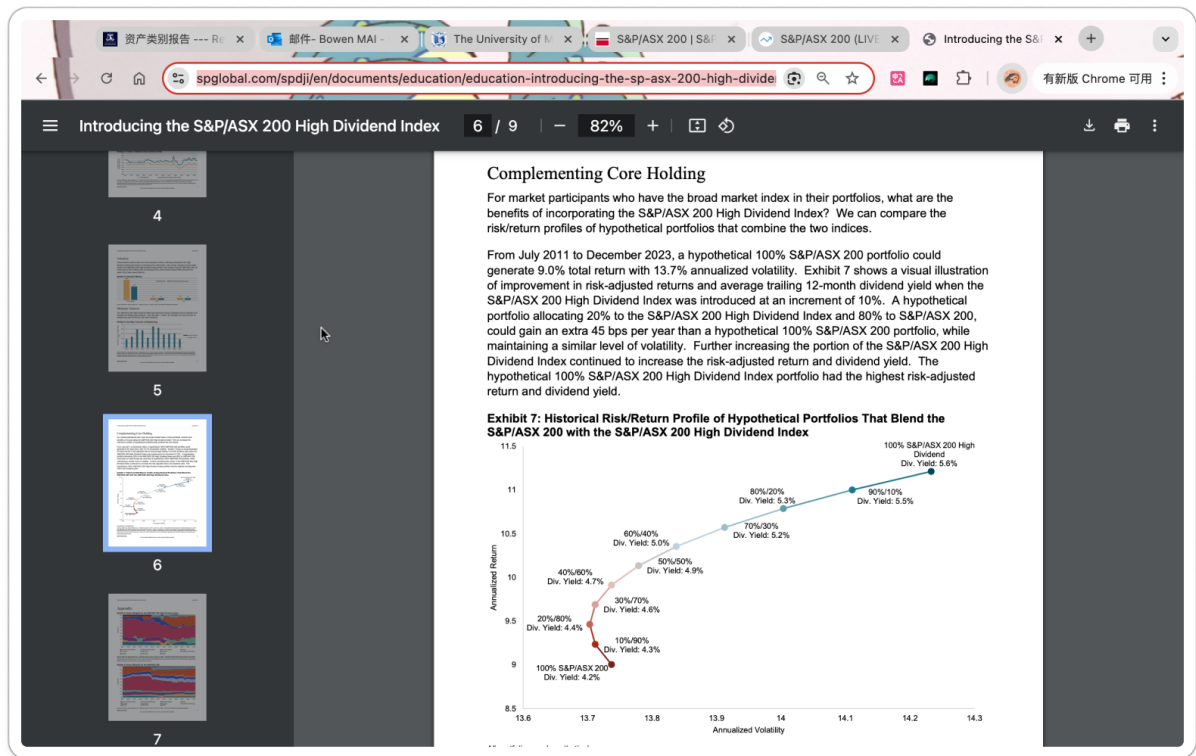
Reference:

ASX 200

Market Index. (n.d.). *S&P/ASX 200 Net Total Return (XNT)*. Retrieved May 8, 2025, from <https://www.marketindex.com.au/asx/xnt>

S&P Dow Jones Indices. (2024, April). *Introducing the S&P/ASX 200 High Dividend Index*. Retrieved May 8, 2025, from <https://www.spglobal.com/spdji/en/documents/education/education-introducing-the-sp-asx-200-high-dividend-index.pdf>





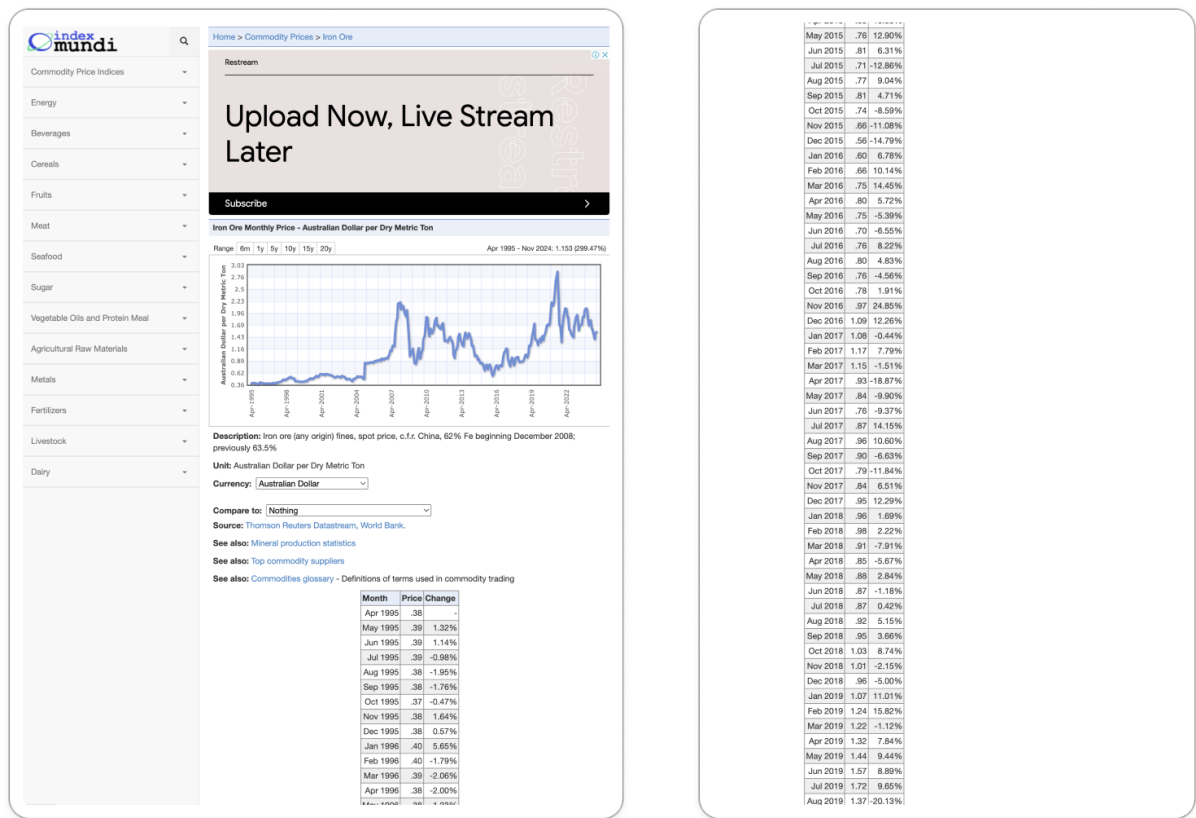
Gold (AUD)

GoldPrice.org. (n.d.). *Gold Price Australia*. Retrieved May 8, 2025, from <https://goldprice.org/gold-price-australia.html>



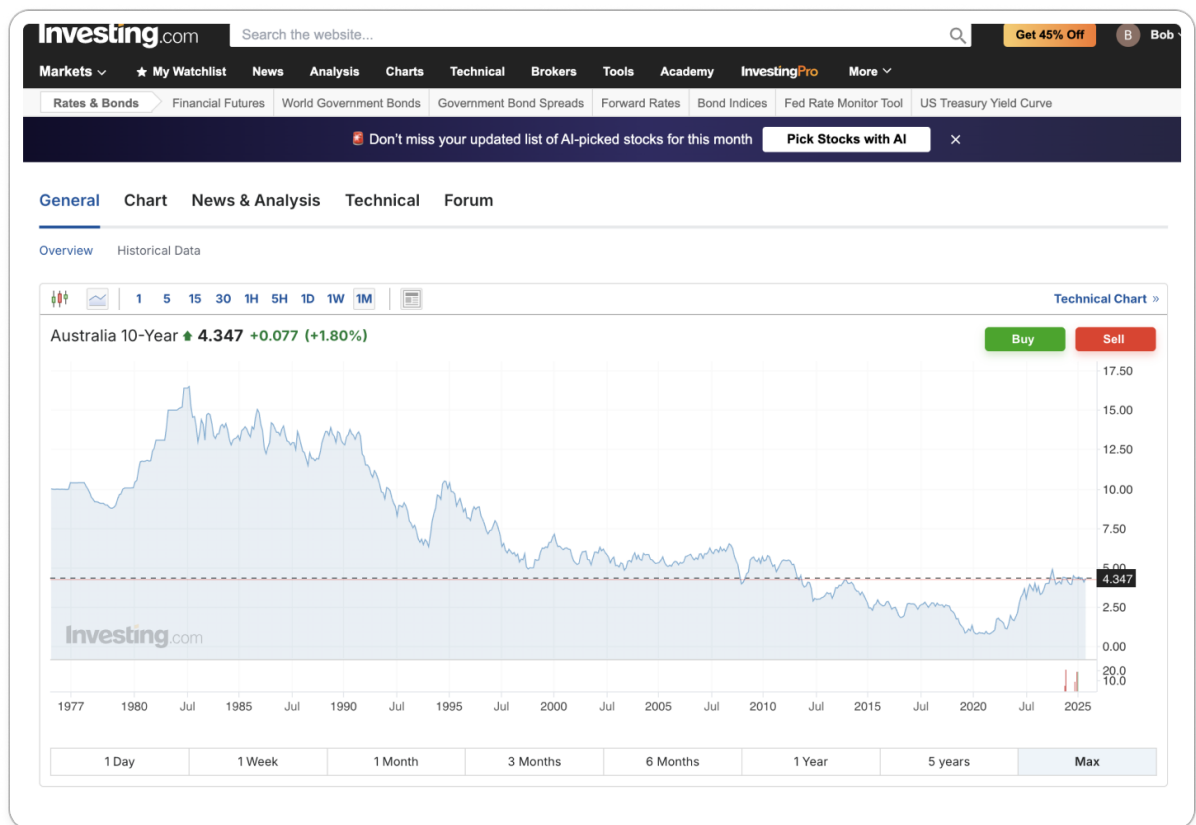
Iron (AUD)

IndexMundi. (n.d.). *Iron Ore Monthly Price - Australian Dollar per Dry Metric Ton*. Retrieved May 8, 2025, from <https://www.indexmundi.com/commodities/?commodity=iron-ore¤cy=aud>



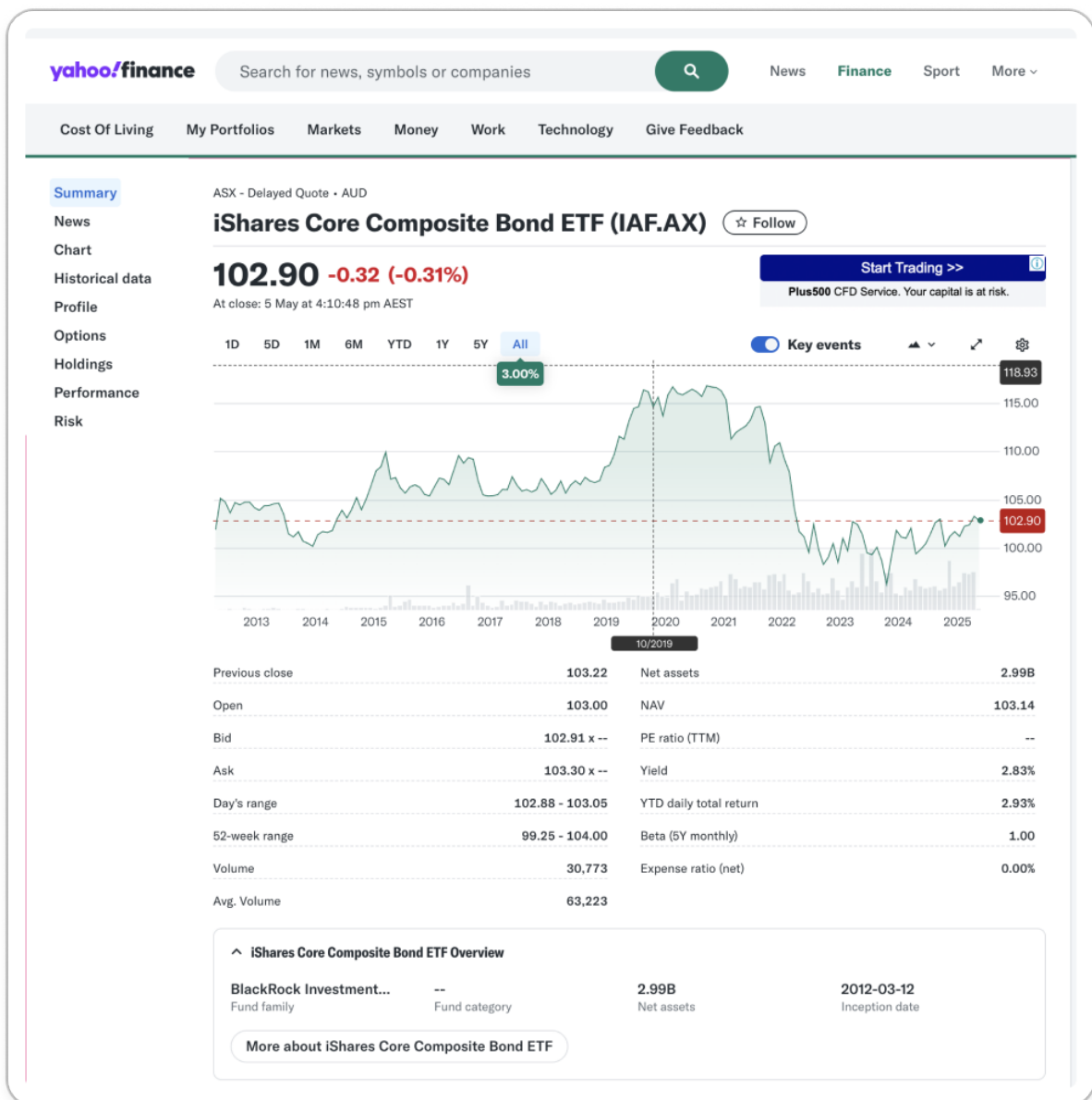
Government Bonds

Investing.com. (n.d.). *Australia 10-Year Bond Yield Historical Data*. Retrieved May 8, 2025, from <https://www.investing.com/rates-bonds/australia-10-year-bond-yield-historical-data>



Corporate Bonds

Yahoo Finance. (n.d.). *iShares Core Composite Bond ETF (IAF.AX)*. Retrieved May 8, 2025, from <https://au.finance.yahoo.com/quote/IAF.AX/>



Residence Property

CoreLogic. (n.d.). *CoreLogic Indices*. Retrieved May 8, 2025, from https://www.corelogic.com.au/our-data/corelogic-indices?utm_source=chatgpt.com



Commercial Office

Yahoo Finance. (n.d.). *DEXUS (DXS.AX) Stock Historical Prices & Data*. Retrieved May 8, 2025, from <https://finance.yahoo.com/quote/DXS.AX/history/>

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Date	Open	High	Low	Close Q	A\$ Close Q	Volume
6 May 2025	7.47	7.51	7.46	7.49	7.49	95,626
5 May 2025	7.53	7.59	7.45	7.49	7.49	1,783,237
2 May 2025	7.50	7.59	7.47	7.51	7.51	2,698,536
1 May 2025	7.46	7.53	7.43	7.52	7.52	2,816,888
30 Apr 2025	7.38	7.52	7.35	7.52	7.52	5,917,864
29 Apr 2025	7.33	7.39	7.28	7.37	7.37	2,747,360
28 Apr 2025	7.27	7.30	7.22	7.28	7.28	3,429,574
24 Apr 2025	7.18	7.26	7.16	7.24	7.24	2,167,874
23 Apr 2025	7.29	7.37	7.22	7.22	7.22	3,132,896
22 Apr 2025	7.15	7.22	7.11	7.16	7.16	2,091,202
17 Apr 2025	7.22	7.24	7.11	7.23	7.23	1,959,819
16 Apr 2025	7.20	7.29	7.17	7.20	7.20	4,047,778
15 Apr 2025	7.15	7.19	7.09	7.12	7.12	2,027,081
14 Apr 2025	7.30	7.16	7.04	7.11	7.11	2,245,275
11 Apr 2025	6.96	7.08	6.93	7.02	7.02	5,286,294
10 Apr 2025	7.29	7.29	7.13	7.15	7.15	4,029,692
9 Apr 2025	6.93	6.98	6.84	6.88	6.88	3,422,575
8 Apr 2025	6.90	7.06	6.86	7.04	7.04	3,990,450
7 Apr 2025	6.72	6.97	6.71	6.96	6.96	6,007,669
4 Apr 2025	7.05	7.13	7.00	7.09	7.09	2,859,505

Industrial Warehouse

Yahoo Finance. (n.d.). *Goodman Group (GMG.AX) Financials*. Retrieved May 8, 2025, from <https://au.finance.yahoo.com/quote/GMG.AX/financials/>

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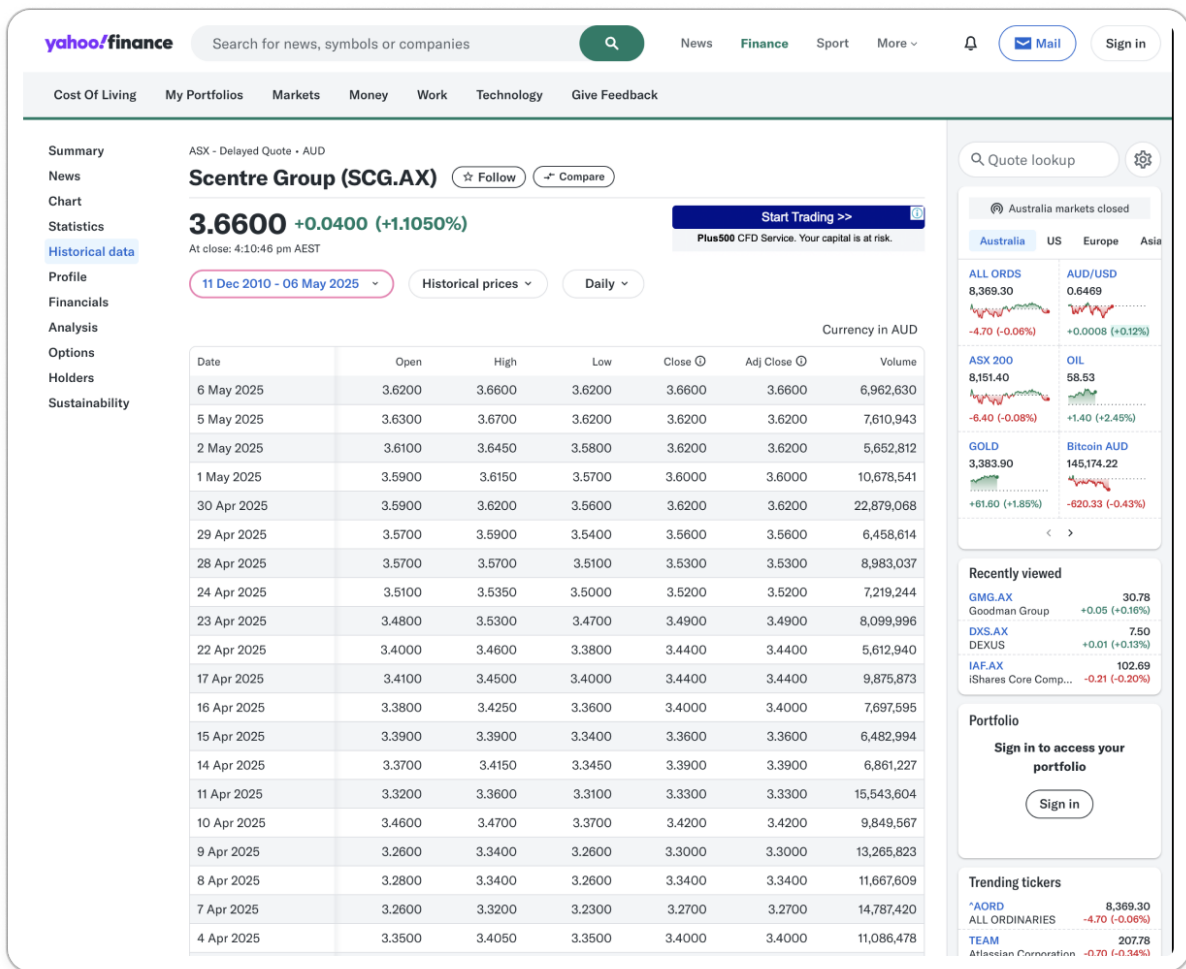
All numbers in thousands

Annual Quarterly ↗ Expand all

Breakdown	TTM	30/06/2024	30/06/2023	30/06/2022
✓ Cash flows from used in oper...	1,188,600	1,188,600	1,284,200	841,000
✓ Investing cash flow	-688,200	-688,200	-716,000	-1,001,500
✓ Financing cash flow	-52,400	-52,400	-308,800	299,700
^ End cash position	1,785,300	1,785,300	1,360,100	1,056,000
Changes in cash	448,000	448,000	259,400	139,200
Effect of exchange rate changes	-22,800	-22,800	44,700	-3,600
Beginning cash position	1,360,100	1,360,100	1,056,000	920,400
Income tax paid supplemental d...	124,600	124,600	164,100	110,500
Interest paid supplemental data	80,200	80,200	59,200	44,200
Capital expenditure	-10,500	-10,500	-13,000	-5,900
Issuance of capital stock	70,300	70,300	67,200	109,700
Issuance of debt	849,800	849,800	1,029,300	1,577,900
Repayment of debt	-456,000	-456,000	-772,000	-789,300
Free cash flow	1,178,100	1,178,100	1,271,200	835,100

Retail Property

Yahoo Finance. (n.d.). *Scentre Group (SCG.AX) Historical Data*. Retrieved May 8, 2025, from <https://au.finance.yahoo.com/quote/SCG.AX/history/?period1=1292108400&period2=1746534116>



Logistics Facilities

Yahoo Finance. (n.d.). *Charter Hall Group (CHC.AX) Historical Data*. Retrieved May 8, 2025, from <https://au.finance.yahoo.com/quote/CHC.AX/history/>

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18.05 +0.49 (+2.79%)

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09 June 2005 - 06 May 2025 Historical prices Daily

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Currency in AUD

Date	Open	High	Low	Close	Adj Close	Volume
6 May 2025	17.49	18.10	17.40	18.05	18.05	1,222,885
5 May 2025	17.56	17.92	17.48	17.56	17.56	793,735
2 May 2025	17.32	17.57	17.11	17.57	17.57	1,082,049
1 May 2025	16.90	17.46	16.90	17.40	17.40	1,124,143
30 Apr 2025	16.77	16.91	16.68	16.90	16.90	2,721,283
29 Apr 2025	16.70	16.86	16.56	16.77	16.77	796,261
28 Apr 2025	16.72	16.74	16.46	16.56	16.56	1,346,013
24 Apr 2025	16.50	16.61	16.36	16.60	16.60	964,707
23 Apr 2025	16.30	16.56	16.21	16.40	16.40	1,049,376
22 Apr 2025	15.96	16.09	15.86	16.05	16.05	551,024
17 Apr 2025	16.18	16.24	15.98	16.24	16.24	485,249
16 Apr 2025	16.02	16.38	15.97	16.13	16.13	1,206,795
15 Apr 2025	16.14	16.23	15.94	16.03	16.03	547,110
14 Apr 2025	16.01	16.26	15.96	16.06	16.06	649,116
11 Apr 2025	15.80	15.96	15.58	15.88	15.88	580,335
10 Apr 2025	16.45	16.49	15.84	16.00	16.00	1,757,041
9 Apr 2025	15.48	15.57	15.09	15.10	15.10	1,990,479
8 Apr 2025	14.90	15.79	14.87	15.79	15.79	2,744,154
7 Apr 2025	14.56	15.15	14.24	15.06	15.06	2,780,106

Quote lookup

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Australia markets closed

Australia US Europe Asia

ALL ORDS8,369.30-4.70 (-0.06%)

AUD/USD0.6468+0.0007 (+0.10%)

ASX 2008,151.40-6.40 (-0.08%)

OIL58.56+1.43 (+2.50%)

GOLD3,393.40+71.10 (+2.14%)

Bitcoin AUD145,168.06-896.44 (-0.61%)

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IAF.AX102.69iShares Core Comp...-0.21 (-0.20%)

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